Transportation Research Part C 92 (2018) 58–75

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

Geometric matching and spatial pricing in ride-sourcing markets

Liteng Zha[a], Yafeng Yin[b,*], Zhengtian Xu[b]

[a] Department of Civil and Coastal Engineering, University of Florida, 365 Weil Hall, Gainesville, FL, United States
[b] Department of Civil and Environmental Engineering, University of Michigan, 2350 Hayward Street, Ann Arbor, MI, United States

ARTICLE INFO

Keywords:
Ride sourcing
Spatial pricing
Geometric matching
Spatial equilibrium and regulation

ABSTRACT

This paper develops a model to investigate the effects of spatial pricing on ride-sourcing markets. The model is built upon a discrete time geometric matching framework that matches customers with drivers nearby. We demonstrate that a customer may be matched to a distant vehicle when demand surges, yielding an inefficient supply state. We further investigate market equilibrium under spatial pricing assuming a revenue maximizing platform, and find that the platform may resort to relatively higher price to avoid the inefficient supply state if spatial price differentiation is not allowed. Although spatial pricing facilitates market clearing, the platform may still set price more than the efficient level, which compromises the public interest. We then propose a commission rate cap regulation that reaps the flexibility of spatial pricing and can achieve the second best under some homogeneity assumptions.

1. Introduction

Ride-sourcing companies such as Uber, Lyft and Didi Chuxing are transforming the way people travel in cities. These companies provide real-time, internet-based platforms that intelligently source ride requests to affiliated private drivers nearby.[1] Compared to the traditional dispatch taxi service, the online matching does not involve a human dispatcher and makes it possible for dispatch to be optimized at scale. The online matching also differs from street hailing in the sense that the "matching" distance between customers and drivers is now enlarged to a few miles, allowing a smaller "critical mass" of customers to sustain the market. Consequently, cities where street hailing was previously unstainable are experiencing a boom in ride-sourcing services.

Ride sourcing employs a similar trip fare structure as traditional taxi services, which includes a frag-drop fee, a time-based and a distance-based charge. For each completed trip, the platform charges a commission that is usually 20–25% of the fare paid by the rider. One of the most important features of the ride-sourcing market is its price differentiation, e.g., Uber's surge pricing and Lyft's prime time pricing. The differentiation is both temporal and spatial. It is advertised to be triggered when the vehicle supply is in shortage (or equivalently if demand surges) so as to attract more drivers to come online or serve the supply-shortage areas as well as limit the service to customers with the higher willingness to pay (Hall et al., 2015; Chen and Sheldon, 2016). Typically, surge pricing is implemented by multiplying a price multiplier to the base trip fare. The price multipliers are area-specific and updated periodically. Both customers and drivers are informed of the price multipliers before transactions occur (Chen et al., 2015). However, the industry practice of surge pricing has led to objection among its customers. Some are frustrated and even angered after being charged significantly higher fares (Curley, 2014; Didi Chuxing, 2017). Others expressed concern that platforms use surge pricing to collect

* Corresponding author.
E-mail address: yafeng@umich.edu (Y. Yin).

[1] The business models of these companies are becoming more comprehensive, and their service features include ridesharing and on-demand shuttles, etc.

https://doi.org/10.1016/j.trc.2018.04.015
Received 18 December 2017; Received in revised form 14 April 2018; Accepted 18 April 2018
Available online 01 May 2018
0968-090X/ © 2018 Elsevier Ltd. All rights reserved.

more revenue under the current percentage commission structure. The public outcry is further intensified when one single platform dominates the marketplace. To respond, Didi has self-imposed a dual cap control on price surge since March 2017[2] (Didi Chuxing, 2017).

In contrast to the fast growth of the ride-sourcing industry, the reaction from regulatory agencies has somewhat lagged. The current efforts still focus on ensuring its level of service such as vehicle inspection, insurance requirement and driver censoring. To our best knowledge, there is no formal regulation on the platform's price setting. This study contributes to the ongoing quest of developing regulatory policies for ride-sourcing services. We limit our scope to the spatial price differentiation practice and examine its impact on the imbalance between labor supply and customer demand across various regions. We attempt to demonstrate spatial pricing as an effective strategy for the platforms to mitigate the supply–demand imbalance and facilitate market clearing. To protect the public interest, we further explore a simple regulation policy for surge pricing without comprising too much its flexibility.

Our model is built on a discrete-time geometric matching framework where a customer is matched to her closest available vehicle within a certain radius. We explicitly model the frictions in the matching and meeting process. Specifically, both customers and drivers will spend time on waiting to be matched. The matching time at each side increases with the number of agents on the same side due to the competition in receiving or providing services. In addition, the matched pair will wait for another en route time before a vehicle is physically occupied (Castillo et al., 2017; Xu et al., 2017). We analytically identify the tipping point between less efficient system states and more efficient ones (Castillo et al., 2017). The tipping point can be judged by examining the ratio between the intensity of unmatched or idle vehicles and that of the total unoccupied (idle plus en route) vehicles in a geographic area, providing a remarkably easy way of signaling the imbalance between supply and demand in practice.

We then proceed to investigate spatial pricing assuming a revenue-maximizing platform and considering both the short- and long-run labor supply. We analytically characterize the optimal price formula and numerically investigate the market states and agents' welfare under both spatial and flat pricing. We find that a platform using only flat pricing may prefer to set a higher price to avoid the supply shortage, especially when facing the uncertainties in marketplace. Spatial pricing is more flexible and in general benefits both the platform and drivers. When price is adjusted below its flat counterpart, customers are also better off. However, a revenue-maximizing platform may set the price above the efficient level. Indeed, we observe that customers are worse off under exceptionally high price surge even though the region is not in shortage of vehicle supply.

Considering the ambiguity of spatial pricing on the welfare of customers, we propose a commission rate regulation to protect the customers, which caps the amount of commission the platform can receive from each unit of service time (or distance). It aims to align the platform's objective of revenue maximization with that of maximizing occupied vehicle hours, which generally benefits customers. The additional revenue from the price surge completely goes to drivers. With a homogeneity assumption on agents' valuation, we show that such a simple regulation can achieve an idealized second best. We note that Didi's self-imposed control on surge pricing mirrors the proposed regulation, although theoretically we do not suggest a ceiling on price multipliers as Didi does (Didi Chuxing, 2017).

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the specifications of demand, supply and geometric matching while the market equilibrium under spatial pricing is discussed in Section 4. Section 5 introduces our empirical study of spatial pricing using data from Didi Chuxing and Section 6 discusses its outcomes. Subsequently, Section 7 investigates the regulation policy. Lastly, we conclude this paper and point out future research needs in Section 8.

2. Literature review

Given an increasing number of studies on ride sourcing, below we present a review of literature relevant to pricing of the services.

A few studies conducted by Uber explore how labor supply responds to surge pricing (Hall and Krueger, 2015; Chen and Sheldon, 2016). They find that drivers will work longer given an unexpected price surge. Based on a large-scale dataset from Uber's API, Chen et al. (2015) demonstrate that the dominant effect of surge pricing is to suppress demand. There is no clear evidence that surge pricing induces drivers to highly surged zones at least in the short time interval during the surge. Their finding seems to be confirmed in an experimental study by Lyft (Nicholas, 2016).

Other studies aim to evaluate the performance of surge pricing and its welfare impacts on market agents. Banerjee et al. (2015) approach this problem using an M/M(k)/1 queuing model. They model surge pricing as a threshold-based dynamic pricing that depends exclusively on the number of idle vehicles. They show that surge pricing outperforms the optimal static pricing when the platform has limited information. Zha et al. (2017) investigate the time-of-day welfare effects of surge pricing. A multi-period time-extended model is introduced to capture drivers' work scheduling decisions. They find that surge pricing benefits the platform and drivers while the effect on customers is mixed. The potential harm to customers is also confirmed in a study by Guda and Subramanian (2017) where a two-zone, two-period model is developed to understand the temporal and spatial effects of surge pricing. They show that the platform may strategically increase price to stifle demand in locations of sufficient supply in order to induce drivers to move to areas where supply is in shortage. Then, price surge can even hurt the welfare of each market player. However, they highlight that surge pricing can send trustfully signal to drivers that there is a great need to move. Most recently, Galichon and Hsieh (2017) analyze surge pricing utilizing a centralized matching framework. They demonstrate the key fact that price is transferable among agents but not the waiting cost. Without price surge, customers and drivers may incur extended waiting

[2] Didi limits the price multiplier to 1.5 and the amount of surge from each trip to CNY 29 for Didi Express, and CNY 59 for Didi Premium. In addition, Didi has committed the additional commission from the surge to drivers. Therefore, the revenue Didi claims is independent of its surge pricing (Didi Chuxing, 2017).

cost, a source of efficiency loss. A revised surge pricing algorithm is proposed to minimize the total market frictions considering demand and supply uncertainties.

Another thread of research looks into the fixed percentage commission adopted by major ride-sourcing companies and attempts to investigate how well this commission structure performs in the context of surge pricing. Cachon et al. (2015) study the optimal compensation contract using a two-period newsvendor-type model. They find that surge pricing generally works well with the fixed percentage commission and yields results close to the optimal contract where the platform dynamically determines both the price (for the customers) and wage (for the drivers). Following the same model setting as Cachon et al. (2015), Hu and Zhou (2017) conclude that the fixed percentage commission can achieve at least 3/4 of the maximal profit under mild assumptions.

Among all previous efforts, two studies are most relevant to this work, namely, Bimpikis et al. (2016) and Castillo et al. (2017). We impose a similar assumption as Bimpikis et al. (2016) to describe idle car drivers' zonal choice. As our modeling focus is on geographic matching, for analytical convenience we do not trace the movement of idle vehicles. By doing so, Bimpikis et al. (2016) investigate how the asymmetric nature of the demand pattern affects the optimal spatial pricing in a network. Despite the insightful discussions therein, the spatial equilibrium in their setting is indeterminate, i.e., the number of variables is more than the number of constraints, and thus the outcomes under optimal pricing may best seen as an ideal bound.

Castillo et al. (2017) identify a matching failure in the ride-sourcing market that an idle driver is matched with a distant customer and has to waste substantial time on picking her up. They show that given a fixed number of working drivers in the system, such a matching failure will cause the trip supply curve to be backward bending. In this backward-bending regime, the ride-sourcing system operates inefficiently. They then suggest that surge pricing can suppress the demand and thus allow the system to stay away from this regime. Our work differs from Castillo et al. (2017) in the following aspects. First, their model is based on an aggregate market while ours considers a spatially differentiated market. Subsequently, the variations in supply and demand in their model come from different times of day while in our framework, these variations result from the heterogeneity in spatial regions. Secondly, Castillo et al. (2017) consider a continuous-time matching process without a matching radius while we assume a discrete-time process with a matching radius. Subsequently, our derivations of customers' waiting time differ. Castillo et al. (2017) assume zero time for a customer waiting for being matched with an idle driver, which may not be the case especially when supply is in shortage and a customer may spend several minutes on waiting to be matched. In contrast, we explicitly consider customer's waiting time for matching in addition to the en route time after a successful matching. Lastly, we consider a revenue-maximizing ride-sourcing platform and then discuss a simple regulation policy to protect the public interest, which is not in the scope of Castillo et al. (2017).

3. Model

This section presents our model of the ride-sourcing market. For simplicity, we only consider a single ride-sourcing platform, and each vehicle can only pick up one customer during a trip. Given an urban region divided into different geographic zones, we assume that the market condition in each zone is homogeneous. We will first specify demand and supply, and then model the platform's matching between customers and drivers from a geometric probability perspective.

3.1. Demand

It is well recognized that the ride-for-hire market is not only cleared by price, since each customer has to wait for some time before being served (e.g., Douglas, 1972; Yang et al., 2010a,b; Taylor, 2016; Castillo et al., 2017). We define a customer's waiting time as the sum of the matching time for a ride to be confirmed plus the en route time for pick up. Consider the average hourly demand at each zone. Let F_i and w_i^c be the average trip fare and customers' average waiting time, respectively. The average demand rate Q_i is assumed to be a decreasing function of the average total cost:

$$Q_i = f(F_i + \beta w_i^c), \quad \forall\, i \in I \tag{1}$$

where β is the customers' value of time and $f' < 0$ by assumption. Without loss of generality, we normalize the flag-drop fee to zero and only consider the time-based charge. Therefore, $F_i = \gamma_i \omega l_i$ where γ_i is the price multiplier; ω represents the charge per unit time and l_i is the average trip duration. We do not consider traffic congestion and l_i is thus treated as a constant. The price multiplier is the same for all customers in zone i.

3.2. Supply

For transporting a customer from zone i, a driver obtains $(1-\eta)$ of the trip fare while the remaining will be the commission to the ride-sourcing platform. We consider two extreme cases on how drivers may respond to the average wage rate, namely, (i) a perfectly elastic labor supply and (ii) a perfectly inelastic labor supply. In the former, we assume that the size of driver pool will be adjusted such that the average wage rate is equal to drivers' hourly opportunity cost c, i.e.,

$$(1-\eta) \sum_i F_i Q_i / \sum_i N_i = c, \quad \forall\, i \in I \tag{2}$$

where N_i is the average number of drivers in zone i. This characterization is more appropriate in the long run where drivers are flexible to make a career choice. In the latter, the total fleet size $N = \sum_i N_i$ is fixed, independent of drivers' wage rate. This is a more proper characterization of labor supply in the short run.

All drivers are assumed to work for the entire modeling period, e.g., one hour, and the variation in total labor supply is due to the change in the number of drivers. This, however, overlooks the variation in drivers' work scheduling that includes start, end time and effective work hours (Zha et al., 2017). In fact, a growing body of literature in labor economics is dedicated to better understanding drivers' work hour choice preferences (e.g., Farber, 2015; Chen and Sheldon, 2016; Vincent, 2016).

3.3. Geometric matching

3.3.1. Technology

The matchings between customers and drivers are completed algorithmically by the ride-sourcing platform. We simplify the algorithm to a discrete time geometric matching process that matches a requesting customer to her closest vehicle within a coverage radius r (Ranney et al., 2015).[3] We assume that customers or drivers will not cancel the transaction once a match is made. We further require that the area of each zone is sufficiently large compared to the matching radius (one can always merge zones to satisfy this requirement if necessary), and thus assume both the customer and driver in a successful matching come from the same zone.

To delineate the matching process, we discretize one analysis period into smaller time steps of equal length δ. At each time step, we consider the following process for each spatial zone:

P1: At the beginning of each time step, all unmatched and newly arrived customers send their requests to the platform and remain stationary at their current locations.
P2: The platform *randomly* loops through all requesting customers located in the region. For each customer, it checks whether a vacant vehicle is available within distance r from the customer. If so, the platform assigns the closest vehicle to her and the matched pair disappears from the network; otherwise, it does nothing.

Note that randomly selecting a customer for consideration of matching at P2 is an approximation to some platforms' practice of matching customers in a first-come-first-serve (FCFS) manner (Castillo et al., 2017).[4] As customers' request sequence is independent of their locations in a spatial realization, the platform operates as if it were selecting a customer at random from the requesting customer pool in the zone.

As previously mentioned, the average waiting time consists of two parts:

$$w_i^c = w_i^{cm} + m_i\ , \quad \forall\ i \in I \tag{3}$$

where w_i^{cm} is the average matching time, or more specifically, the expected time for a requesting customer to be matched, while m_i represents the average en route time, i.e., the expected travel time for the driver to pick up the matched customer after a successful matching. By symmetry, drivers' average waiting or search time until picking up a customer will be:

$$w_i^v = w_i^{vm} + m_i\ , \quad \forall\ i \in I \tag{4}$$

where w_i^{vm} is the expected time for a vacant vehicle to be matched. As the side in shortage will be matched almost instantly, we have:

$$\delta \approx \min\{w_i^{vm}, w_i^{cm}\} \tag{5}$$

where, as aforementioned, δ is the length of a time step, a modeling parameter to be estimated in Section 5. For analytical tractability, a continuous approximation of Eq. (5) yields:

$$w_i^{vm} \approx -w_i^{cm}\ln(1-\delta/w_i^{cm})\ , \quad \forall\ i \in I \tag{6}$$

which implies that $\lim_{w_i^{cm}\to\infty} w_i^{vm} = \delta$; $\lim_{w_i^{vm}\to\infty} w_i^{cm} = \delta$. It can be verified that w_i^{vm} is convex and decreasing in w_i^{cm}. A sample plot is given in Fig. 1.

We denote the intensities of unmatched customers and vehicles at the beginning of each time step as N_i^{cm} and N_i^{vm} $\forall\ i \in I$, respectively. At the steady state, the following relationship holds:

$$\begin{aligned} N_i^{vm} &= Q_i w_i^{vm}/A_i\ , \quad \forall\ i \in I \\ N_i^{cm} &= Q_i w_i^{cm}/A_i\ , \quad \forall\ i \in I \end{aligned} \tag{7}$$

where A_i is the area of zone i. Then Eq. (6) can be rewritten as:

$$w_i^{cm} = \frac{\delta}{1-\exp(-N_i^{vm}/N_i^{cm})}\ , \quad \forall\ i \in I \tag{8}$$

The denominator can be interpreted as the probability of a customer being matched in one step. Then her average matching time is the duration of a time step δ times the average number of steps she is going to wait.

We denote by x the distance of the closest unmatched vehicle to an arbitrary customer. If we assume that the arrivals of

[3] The cap on r is not a limit from the matching technology but due to agents' behavioral response, e.g., a driver matched to a customer ten miles away is likely to reject the order. As mentioned in Castillo et al. (2017), Uber sets a cap of 20 min of en route time, beyond which the system will cancel the order.

[4] The FCFS rule aims to minimize customers' average matching time. Castillo et al. (2017) further discuss several operational reasons behind the rule. A different matching algorithm by paring mutually closest customers and drivers is studied in Xu et al. (2017).

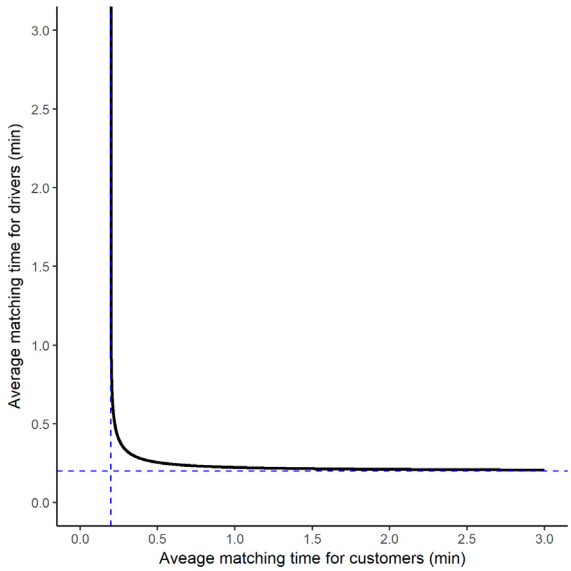


Fig. 1. Average matching times for drivers and customers with δ equal to 12 s.

unmatched vehicles follow the spatial Poisson point process, the probability density function of x simply becomes (e.g., Moltchanov, 2012; Chiu et al., 2013):

$$g(x) = 2\pi x N_i^{vm} e^{-\pi x^2 N_i^{vm}}, \quad x \geqslant 0\ , \quad \forall\ i \in I \tag{9}$$

This leads to the expected travel time between an arbitrary chosen customer and her closest unmatched vehicle:

$$\begin{aligned} E_i &= \frac{k}{v}\int_0^r x g(x|x \leqslant r)dx \\ &= \left[-\frac{kr}{v}e^{-\pi r^2 N_i^{vm}} + \frac{k}{2v\sqrt{N_i^{vm}}} erf\left(\sqrt{\pi N_i^{vm}}\, r\right)\right] \Big/ \left(1 - e^{-\pi r^2 N_i^{vm}}\right), \qquad \forall\ \ i \in I \end{aligned} \tag{10}$$

where $k \approx \sqrt{\pi/2}$ is a scaling parameter that adjusts the Euclidean distance to the Manhattan distance (Arnott, 1996); v is the average speed of the vehicle and $erf(x) = 2/\sqrt{\pi}\int_0^x e^{-t^2}dt$ is the Gauss error function. When $\pi r^2 N_i^{vm} \geqslant 2$, Eq. (10) can be approximated by:

$$E_i \approx \frac{k}{2v\sqrt{N_i^{vm}}}\ , \quad \forall\ i \in I \tag{11}$$

Eq. (11) prescribes the average en route time when $N_i^{\mathrm{vm}} \gg N_i^{\mathrm{cm}}$. Otherwise, using it as the average en route time is biased downwards. In fact, the number of unmatched vehicles decreases as the platform loops through customers (P2). Customers matched later in the matching step will incur longer en route time. Generally, it is not accurate to assume that each customer experiences the same N_i^{vm}. We assume that the average en route time takes the following form:

$$m_i = \frac{k}{2v(N_i^{vm})^{\alpha}}\ , \quad \forall\ i \in I \tag{12}$$

where α is expected to vary in (0,0.5].

3.3.2. Multiple supply states

Similar to Arnott (1996), Yang et al. (2005) and Castillo et al. (2017), the geometric matching framework depicted in Eqs. (3), (4), (6), (7) and (12) may possess multiple supply states associated with a given ($\boldsymbol{Q}$, $\mathbf{w^v}$). Let $\kappa_1 = \frac{1+\alpha}{\alpha}\left[\frac{k\alpha}{2v(Q_i/A)^{\alpha}}\right]^{\frac{1}{1+\alpha}}$ and $\kappa_2 = \delta + \frac{k}{2v(Q_i\delta/A)^{\alpha}}$. For the given ($\boldsymbol{Q}$, $\mathbf{w^v}$), the following proposition describes the possible number of the supply sates at zone i:

Proposition 1. The supply has (i) no corresponding state if $w_i^v < \kappa_1$; (ii) one state if $w_i^v = \kappa_1$ or $w_i^v \geqslant \kappa_2$ and (iii) two states if $\kappa_1 < w_i^v < \kappa_2$.

The proof is presented in Appendix A[5] and an illustrative plot is given in Fig. 2. It is noted that given Q_i, w_i^v is convex in w_i^{vm} and the minimum of w_i^v is achieved when $w_i^{\mathrm{vm}} = \frac{\alpha}{1+\alpha}w_i^v$. The implications from case (i) and (ii) are straightforward. When w_i^v is extremely small, the number of unmatched vehicles cannot sustain the (arbitrarily) chosen Q_i; on the other hand, w_i^{vm} will account for the

[5] Appendix A contains the proofs of all propositions and lemmas in this paper.

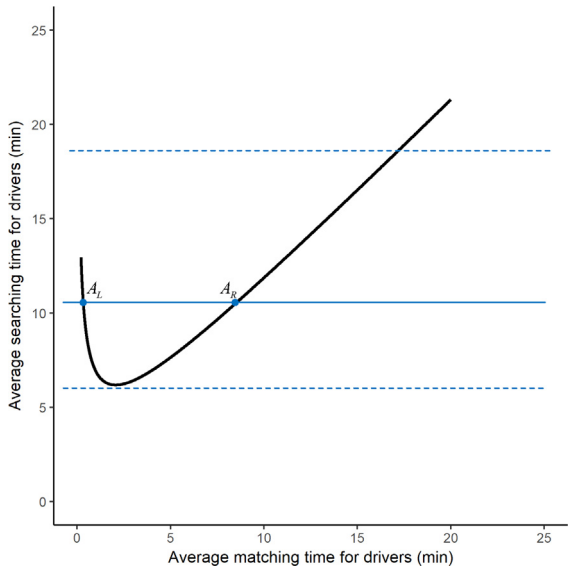


Fig. 2. Demonstration of multiple supply states.

majority of w_i^v when vehicle supply is sufficient. The most interesting part is case (iii) where two states of completely different market conditions can occur.

Let A_L and A_R denote the corresponding supply states in Fig. 2. We note that A_L is characterized by the shortage of idle vehicles while A_R features a marketplace with relatively sufficient number of idle vehicles. Similar to the wild-goose-chase identified by Castillo et al. (2017), A_L represents a situation in which drivers spend a substantial portion of time on en-route trips to pick up customers. The mismatch is due to the practice of the platform to quickly commit a customer with a ride while no idle vehicles are nearby. For a complete comparison between A_L and A_R, we measure the welfare of the customers by consumers' surplus (CS) and that of the platform and drivers by producers' surplus (PS)[6] as follows:

$$\begin{aligned} CS_i &= \int_0^{Q_i} f^{-1}(z)dz - (F_i + \beta w_i^c)Q_i\,, \quad \forall\, i \in I \\ PS_i &= F_i Q_i - cN_i\,, \quad \forall\, i \in I \end{aligned} \tag{13}$$

Then we have the following:

Lemma 1. $CS_{i(A_L)} = CS_{i(A_R)}$ and $PS_{i(A_L)} < PS_{i(A_R)}$.

Since price is transferable between consumers and producers but not the waiting cost (Galichon and Hsieh, 2017), comparing the two states shows that a higher price will reduce customers' waiting cost while maintaining the same level of the generalized cost. This does not hurt the surplus of the customers but increases the joint revenue of the platform and drivers. Hereinafter, we call A_L as dominated state and A_R dominating state. The threshold for state change is $w_i^{\mathrm{vm}} = \frac{\alpha}{1+\alpha} w_i^v$, or in a more identifiable fashion, $N_i^{\mathrm{vm}} = \frac{\alpha}{1+\alpha} N_i^v$: the intensity of idle vehicles is less than $\frac{\alpha}{1+\alpha}$ of the vacant vehicles in a geographic area. The constant term $\frac{\alpha}{1+\alpha}$ is the result of the functional form of the en route time.

For the remaining discussion, we assume that there is at least one equilibrium point from the geometric matching. Rather than referring to Eqs. (3), (4), (6), (7) and (12), the following implicit waiting time formula is introduced to facilitate the presentation:

$$w_i^c = W(Q_i, w_i^v)\,, \quad \forall\, i \in I \tag{14}$$

We further assume that $W(Q_i, w_i^v)$ is continuously differentiable with respect to Q_i and w_i^v. We drop the arguments of $W(\cdot,\cdot)$ whenever it is self-explanatory and define $W_1 = \partial w_i^c / \partial Q_i$ and $W_2 = \partial w_i^c / \partial w_i^v$. Lemma 2 states that the signs of the partial derivatives are affected by the location of the equilibrium solution.

Lemma 2. $W_1 < 0, \quad W_2 < 0$ if the solution point is in the dominating state and $W_1 > 0, \quad W_2 > 0$ if the solution point is in the dominated state.

3.3.3. Comparison to street hailing

The matching technology of ride sourcing is more advanced than street hailing, as the latter is featured by a much smaller matching distance. Information imperfection often arises when unmatched customers (drivers) wish to find (provide) rides but do not know the existence of agents on the other side. Consequently, customers typically spend much longer time on waiting to be

[6] We do not consider any fixed cost, so producers' surplus is the joint profit of the platform and drivers. The platform's cost is further normalized to zero.

"matched". On the other hand, the en route distance is negligible in street hailing, as the customer and the driver in a matched pair will be mutually closest to each other. This implies that the mismatch problem will not occur in street hailing.

The analytic form of customers' average waiting time is given by Douglas (1972):

$$w_{is}^{c} \propto \frac{1}{N_i^{vm}}, \quad \forall\, i \in I \tag{15}$$

By inspecting the form of Eqs. (12) and (15), one can verify that customers choosing street hailing will experience more percentage reduction in the average waiting time compared to those with ride sourcing if the intensities of unmatched vehicles and customers double. That is, street hailing has higher returns to scale than ride sourcing. Although ride sourcing is generally more efficient, the ratio of m_i to w_{is}^{c} increases in the order of $(N_i^{vm})^{\alpha}$. In dense marketplaces such as Manhattan, NY, ride sourcing may not necessarily offer lower average waiting time to customers (Arnott, 1996; Nie, 2017).

4. Equilibrium under spatial pricing

In a spatial network, idle drivers search for customers in zones that they perceive to offer the highest payoff. Assuming that drivers have perfect information on the profitability of each zone, an equilibrium state can be achieved where the distribution of the vacant vehicles across all zones makes each zone equally profitable (Lagos, 2000; Yang et al., 2010a; Bimpikis et al., 2016):

$$(1-\eta)F_i Q_i / N_i = \pi, \quad \forall\, i \in I \tag{16}$$

where π can be interpreted as drivers' wage rate. Note that Eq. (16) coincides with drivers' entry condition in the long run if $\pi = c$. A possible generalization of (16) is via a complementarity formulation discussed in Appendix B. However, the generalization increases the model complexity and makes analytical characterization impossible.

A vehicle is considered to be either occupied or vacant. A vacant vehicle is either idle to be matched or en route to pick up a customer. Then the following conservation condition holds:

$$N_i = Q_i(w_i^{v} + l_i), \quad \forall\, i \in I \tag{17}$$

The vacant vehicle hour ($Q_i w_i^{v}$) consists of the vehicles that have searched in zone i, whether or not they succeed in being matched with customers in the zone, while the corresponding occupied vehicle hour ($Q_i l_i$) accounts for all the matched vehicles departing from zone i. Then, an alternative specification to Eq. (16) becomes:

$$(1-\eta)F_i / (w_i^{v} + l_i) = \pi, \quad \forall\, i \in I \tag{18}$$

where $(1-\eta)F_i$ and $w_i^{v} + l_i$ can be respectively treated as the average reward and the service duration for completing a trip.

Given a spatial pricing scheme, we are now ready to define its market equilibrium under the scheme:

Definition 1. Given trip fare $\mathbf{F}$ and commission percentage η, a steady-state spatial equilibrium is achieved if:

(1) The customer demand satisfies Eq. (1);
(2) The total fleet size N remains fixed in the short run; or adjusted such that Eq. (16) holds in the long run with $\pi = c$;
(3) Given $(\mathbf{Q}, \mathbf{w^v})$, there exists at least one supply state for Eq. (14);
(4) The conservation in vehicle status, i.e., Eq. (17) holds.

Next, we investigate how the platform determines the spatial pricing scheme and the resulting market outcomes. We consider a revenue-maximizing platform and its pricing setting can be described by the following two problems:

(SP-L)

$$\begin{aligned} &\max_{\mathbf{N},\mathbf{z},\eta} = \sum_i \eta F_i Q_i \\ &\text{s.t.} \\ &\quad \pi = c \\ &\quad \text{Eqs.(1),(14),(16),(17)} \end{aligned} \tag{19}$$

(SP-S)

$$\begin{aligned} &\max_{\mathbf{z}} = \sum_i \eta F_i Q_i \\ &\text{s.t.} \\ &\quad \sum_i N_i = N \\ &\quad \text{Eqs.(1),(14),(16),(17)} \end{aligned} \tag{20}$$

where $\mathbf{z} = (\mathbf{w^v}, \mathbf{w^{vm}}, \mathbf{w^c}, \mathbf{w^{cm}}, \mathbf{Q}, \mathbf{F})$. SP-L and SP-S respectively prescribe the optimal pricing scheme and the associated equilibrium in the long run and short run. Both the fleet size and the commission percentage will become endogenous variables in SP-L.

The optimal pricing formula under SP-L is characterized by:

$$F_i = c\rho_{Li}(w_i^v + l_i) + \beta Q_i W_1 - \frac{Q_i}{f_i'}, \quad \forall\, i \in I \tag{21}$$

where $\rho_{Li} = \lambda_i/[\eta + \lambda_i(1-\eta)]$ and λ_i is the Lagrangian multiplier associated with Eq. (16). ρ_{Li} adjusts for the inflexibility of charging a zonal-specific commission. If η can be set freely among zones, SP-L then becomes the aggregation of the independent optimization problem for each zone with $\lambda_i = 1$ and $\rho_{Li} = 1$, $\forall\, i \in I$.

Similar to previous studies (Yang et al., 2010b; Zha et al., 2016), the optimal price contains three components. $c\rho_{Li}(w_i^v + l_i)$ represents the average costs of serving a customer; $-Q_i/f_i'$ is the mark-up term from revenue maximization; and $\beta Q_i W_1$ can be interpreted as the externality from matching. The matching externality implies the total increase of average waiting cost from a unit increase of Q_i while fixing w_i^v at a given level. As implied in Lemma 2, $W_1 < 0$ if the revenue-maximizing point is in the dominating region so that the platform will reduce the charge since the additional increase in Q_i reduces w_i^c (positive matching externality). However, if the revenue-maximizing point is in the dominated region, the platform will charge more as the additional increase in Q_i increases w_i^c (negative matching externality). Proposition 2 below shows that the latter case will not happen:

Proposition 2. The optimal solution from SP-L is in the dominating state.

The optimal pricing formula for SP-S is similar to that of SP-L:

$$F_i = c\rho_{Si}(w_i^v + l_i) + \beta Q_i W_1 - \frac{Q_i}{f_i'}, \quad \forall\, i \in I \tag{22}$$

where $\rho_{Si} = (r\lambda_i + \mu)/[\eta + \lambda_i(1-\eta)]$ and μ is the Lagrangian multiplier associated with Eq. (20). Despite of its similarity to Eq. (21), we are not able to characterize the region of its optimal solution. However, similar to what Arnott (1996) pointed out, if the solution is in the dominated state, it may not be stable.

We denote the corresponding flat pricing problems in the long-run and short-run as FP-L and FP-S. They share the same formulations as SP-L and SP-S with one additional constraint to force a single price multiplier to apply for all zones. The optimal price formulas for FP-L and FP-S are messy, as the platform must strike the balance of all zones due to its inflexibility in price setting. We resort to numerical examples in Section 6 to demonstrate the properties of the optimal solution points.

5. Model estimation

We first introduce a dataset from Didi Chuxing, from which we reconstruct customers' average waiting time. An empirical specification and estimation of the demand function are then discussed.

5.1. Dataset

The original dataset contains 8,540,614 order requests from Jan. 1st -21st, 2016 in an undisclosed Chinese city and was publicly made available by Didi Chuxing (2016a). Didi anonymously divided the study region into 66 zones. We follow their indexing of zones in this paper. Zone 39 is identified as the airport area with an average trip duration of 1.03 h. We filter all trips originating from Zone 39 since the matching mechanism may be different in such a "spot" market. Each request records the customer's ID, driver's ID, trip fare, the timestamp when the request occurs as well as the IDs of the origin and destination. In addition, there are supplementary datasets that contain other information such as weather, PM2.5 indices and the number of community facilities in a zone. After we remove the observations with missing values and duplicate IDs, the master dataset retains 7,816,328 request records. Overall, approximately 83% of the time a customer succeeds in being matched to a driver in one request. Despite the high-resolution order data, some key information is missing, such as the adjacency of the study zones and the waiting time incurred by customers. The latter is reconstructed using a method described next.

5.2. Average waiting time

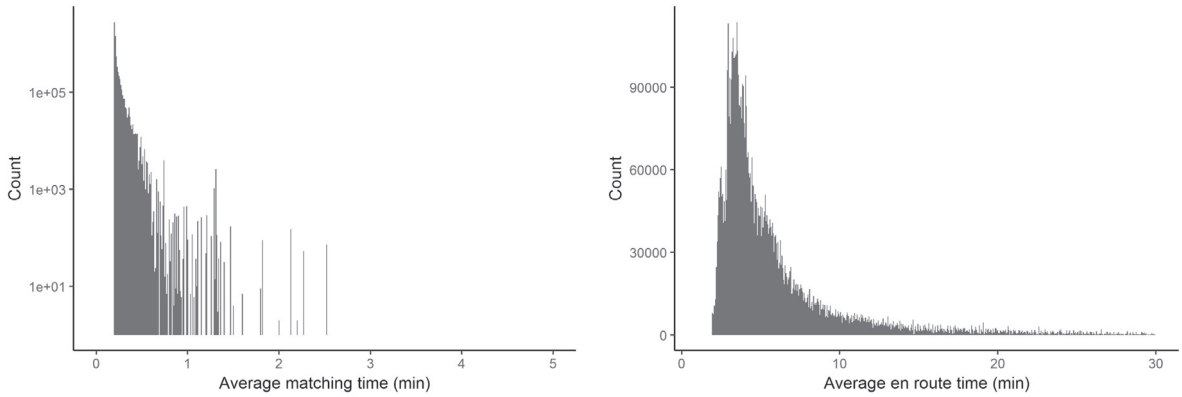
We assume a steady state in each hour. The average values for w_i^{cm}, m_i and w_i^c in an hour are reconstructed utilizing the proposed matching procedure. The process is described as follows:

Step 1. Compute the proportion of requests that are successfully matched, $Pr_i = Q_i/q_i$ where q_i is the rate of requests from zone i. Then, we have

$$N_i^{cm} = \begin{cases} Q_i\delta/(Pr_i A_i), Pr_i > 0 \\ q_i/A_i, Pr_i = 0 \end{cases}, \quad \forall\, i \in I$$

Step 2. Estimate the intensity of the unmatched vehicles given that $Pr_i = 1 - \exp(-N_i^{vm}/N_i^{cm})$:

$$N_i^{vm} = \begin{cases} -N_i^{cm}\ln(1-Pr_i), Pr_i < 1 \\ \psi N_i^{cm}, Pr_i = 1 \end{cases}, \quad \forall\, i \in I$$



(a) Distribution of the average matching time (b) Distribution of the average en route time

Fig. 3. Reconstructed average matching and en route times.

where ψ should be significantly larger than 1 to represent sufficient vehicle supply.

Step 3. Calculate w_i^c, w_i^{cm}, m_i according to Eqs. (3), (8) and (12) when $Pr_i > 0$; otherwise set them to NAs.

The estimation of the matching step duration δ is non-trivial. When vehicle supply is sufficient, customers' requests are fulfilled instantly, subject to approximately two seconds' system response time. We also have consistently observed that customers wait for some time (in a magnitude of minutes) to make another ride request when their previous request was not fulfilled. The lagged times across all realized matchings are approximately 10.2 s. We thus determine δ to be 12 s to account for both the system response time and the lagged times.

We choose $\psi = 10$ and $\alpha = 0.5$ a priori, which of course, may be subject to certain biases. A more appropriate way in determining ψ is to use a high percentile (e.g., 95%) in the distribution of the intensity of unmatched vehicles if more data is available. On the other hand, Castillo et al. (2017) suggest a practical interval for α to be [0.25, 0.5], which is consistent to our theoretical derivation in Section 3.1. The distributions of customers' average matching time and average en route time are given in Fig. 3(a) and (b), with the corresponding mean being 0.23 min and 6.17 min, respectively. The chosen parameters have produced a comparable average en route time to a reported value (Didi Chuxing, 2016b).

5.3. Log-linear demand

We consider a log-linear demand function of the following specification:

$$\log Q_i = \log\theta_0 + \theta_1 F_i + \theta_2 w_i^c + \theta_3 X_i + \varepsilon_i\,, \quad \forall\, i \in I \tag{23}$$

where ε_i is the Gaussian error. We incorporate the number of community facilities X_i in each zone to capture a certain level of spatial heterogeneity (Didi Chuxing, 2016a). It is anticipated that $\theta_1 < 0$, $\theta_2 < 0$ and $\theta_3 > 0$. $\theta_0\exp(\theta_3 X_i)$ is the base demand rate. The ratio of θ_2 over θ_1 can be seen as an estimate of β.

The estimation of demand may be subject to endogeneity problems particularly with the existence of surge pricing. To partially alleviate the endogeneity issues, we focus on a specific period of 7:00–8:00 a.m. during weekdays, which helps constrain some demand shifters. Then, we implement a two-stage least squares (2SLS) approach where F_i is instrumented by the fare in the closest zone to i, i.e., F_{iv} (Hansen, 2017). Note that F_{iv} is positively related to F_i as customers in areas nearby exhibit similar sensitivity to price. On the other hand, F_{iv} is less likely to be related to ε_i that accounts for market dynamics specific to zone i.

The 2SLS estimation is listed in Model III of Table 1. We also report the ordinary least square (OLS) results for another two specifications: without instrumenting trip fares (Model II) and further removing the average waiting time (Model I). The differences

Table 1
Estimation of demand at 7:00 a.m. ~ 8:00 a.m. during weekdays.

Variables	Model I Parameters	S.E.	Model II Parameters	S.E.	Model III Parameters	S.E.
Fare	−0.058	0.006	−0.026	0.004	−0.052	0.009
Waiting time	NA	NA	−2.764	0.099	−2.834	0.270
Intercept	5.239	0.121	5.602	0.090	6.106	0.187
X	2.009	0.086	1.509	0.066	1.298	0.102
R^2	0.489		0.724		NA	
# observations	910		910		910	

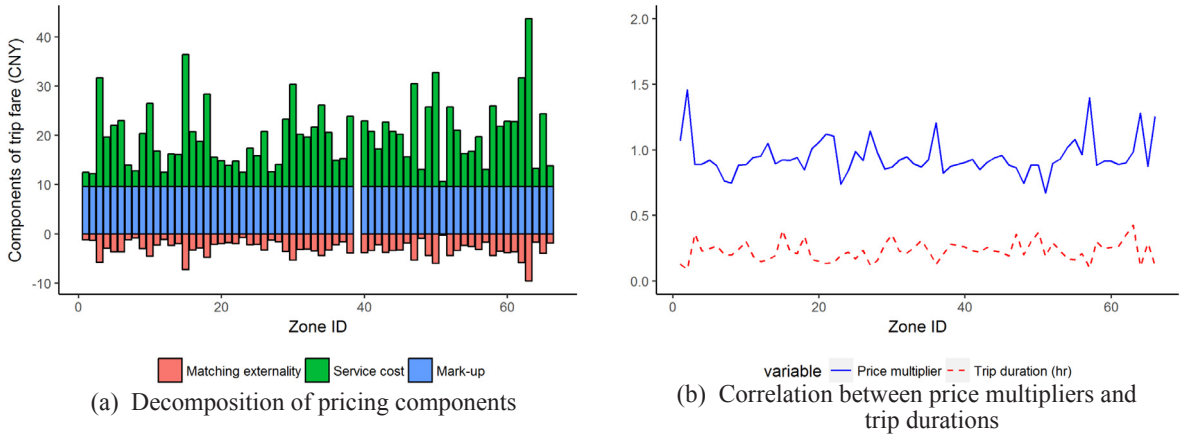


Fig. 4. Optimal pricing under SP-L.

in the magnitude of parameters associated with trip fares (θ_1) between Model I and II imply some level of correlation between F_i and w_i^c. The difference in θ_1 between Model II and III is possibly due to the endogeneity bias. The estimates in Model III are used in the remaining analysis.

Admittedly, the endogeneity issues are not fully addressed in Model III as we haven't found a legitimate instrument for w_i^c. However, the value of θ_2/θ_1 seems to suggest a reasonable value of waiting time of CNY 54.5/h during the peak hour.

6. Numerical experiments

We report the equilibrium results under spatial and flat pricing in this section. We look into the characteristics of the optimal pricing schemes, the region of the equilibrium solutions and the trade-offs in agents' welfare.

6.1. Perfectly elastic labor supply

The estimated parameters in Section 5.3 rely on the data across 21 sampling days and are best seen as the short-run estimates. We thus double the values of θ_1 and θ_2 due to the fact that customers' demand is more elastic in the long term (Castillo et al., 2017). We assume the annual reservation income is 52,935 CNY (Didi Chuxing, 2016b), which gives the equilibrium wage rate approximately CNY 25/h. Didi's current fare structure suggests an hourly rate of 81 CNY assuming a constant speed of 15.6 mile/h for driver partners (Didi Chuxing, 2016b). The parameter α of Eq. (12) is set at 0.5.[7] Accordingly, the threshold for the change between dominated and dominating states is 1/3, implying that when the average matching time over the searching time ratio (MSR) is less than 1/3, the system operates in less efficient states.

The decomposition of the optimal trip fare under SP-L is given in Fig. 4(a). The average service cost and the mark-up charge accounts for the majority of the trip fare. The charge from the matching externality is negative, which confirms that all equilibrium solutions are in the dominating region. From Fig. 4(b), the price multipliers vary from 0.67 to 1.45 for SP-L. If we assume away the spatial variation resulted from $\beta Q_i W_1$ and ρ_{Li}, Eq. (21) suggests that γ_i is opposite to l_i under the log-linear demand function. It seems to be the case from Fig. 4(b) where the correlation coefficient between them is -0.56.

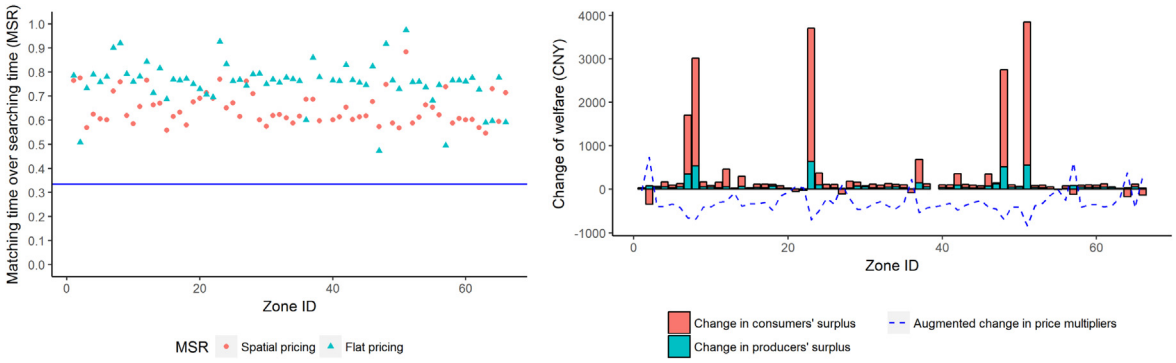
Fig. 5(a) shows that the MSR is above 1/3 for both SP-L and FP-L. Note that the price multiplier for FP-L is 1.09, which exceeds 88% of the price multipliers for SP-L. Due to the inflexibility under flat pricing, the platform has to set relatively higher price multiplier to avoid a market solution being in the dominated region. We plot the difference in the welfare under spatial pricing minus that under flat pricing in Fig. 5(b) and levy the difference in price multipliers (in an augmented scale) by a dashed line. Spatial pricing increases the producers' surplus, but the effect on consumers' surplus is vague. As all market equilibrium solutions are in the dominating region, consumers' surplus decreases in trip fare. In zones where price multipliers are larger than 1.09, we observe that customers are worse off. In contrast, we observe a win–win situation for the remaining zones with lower price multipliers.

6.2. Fixed fleet size

We fix the fleet size at its average value across the study period, which is approximately 7700. We set the commission percentage at 0.2. The original estimates of θ_1 and θ_2 from the third specification in Table 1 are used.

The decomposition of the optimal trip fare under SP-S is given in Fig. 6(a). The charge from the matching externality is still negative, which implies that all the solution points are in dominating region. The matching externality is dominated by the mark-up term alone since we adopt smaller θ_1 (in absolute value). We observe a negative correlation between the price multiplier and trip

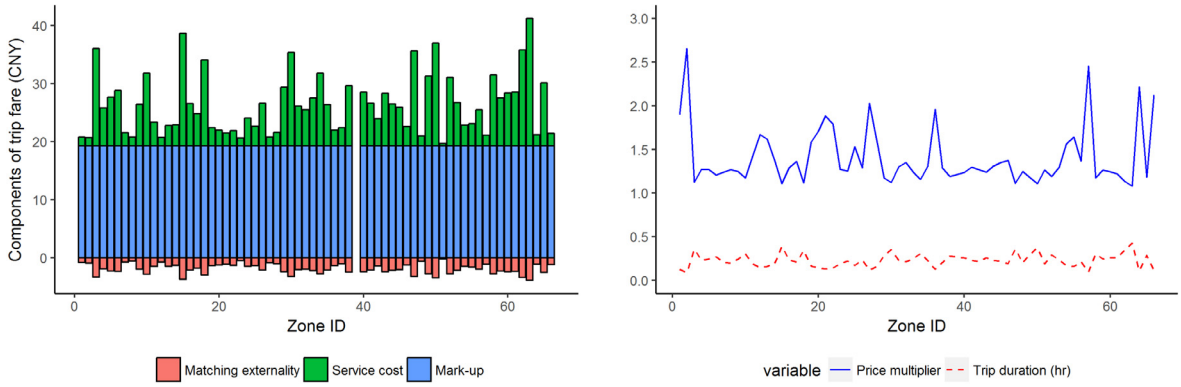
[7] Our observations in this section remain qualitatively unchanged when we experiment with different α values in [0.25, 0.5].



(a) Drivers' average matching time over average searching time under spatial and flat pricing

(b) Display of welfare changes under spatial and flat pricing

Fig. 5. Long-run equilibrium outcomes under spatial and flat pricing.



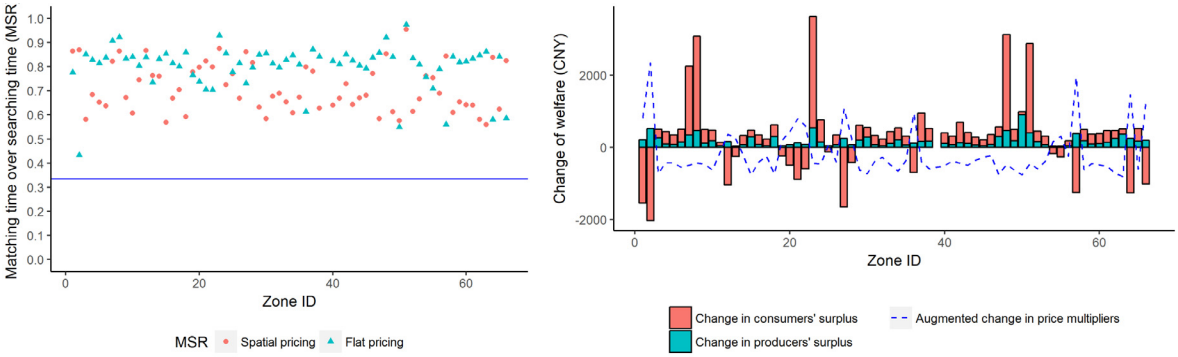
(a) Decomposition of pricing components

(b) Correlation between price multipliers and trip durations

Fig. 6. Optimal pricing under DP-S.

duration in Fig. 6(b) with the correlation coefficient of -0.81.

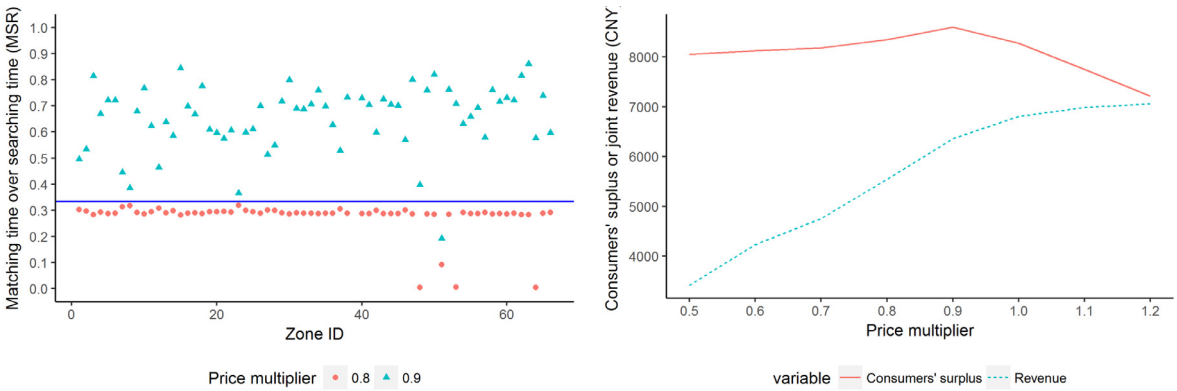
Similarly in Fig. 7, we find that the MSR for all zones is above 1/3 for SP-S and FP-S. Note that the price multiplier for FP-S is 1.49 which is approximately the 76th percentile of the price multipliers for SP-S. SP-S yields an increase in producers' surplus, and again, the change in consumers' surplus is mixed. However, an increasing number of zones experience deteriorating consumers' surplus



(a) Drivers' average matching time over average searching time under spatial and flat pricing

(b) Display of welfare changes under spatial and flat pricing

Fig. 7. Short-run equilibrium outcomes under spatial and flat pricing.



(a) Drivers' average matching time over average searching time

(b) Decomposition of welfare at different price levels

Fig. 8. Short-run market dynamics at different price levels.

under spatial pricing. This is mainly because demand is less elastic in the short run and the platform can set higher prices for revenue maximization.

6.3. Fixed fleet size with suboptimal price multipliers

The optimal pricing discussed previously is at best seen as an approximation to well-anticipated market conditions. However, the marketplace may run into unexpected demand surges. The problem is alleviated if price is adjustable but may not be the case for flat pricing. To mimic the scenario with less informed drivers and platform respectively, we drop the equal profitability constraint and investigate the market outcomes at price levels lower than the optimal one. We still focus on the short-run problem with a fixed fleet size of 7700.

When the price multiplier drops (from the optimal level of 1.25 in the relaxed problem), producers' surplus decreases while consumers' surplus first increases and then decreases as in Fig. 8(b). The tipping point for the change in consumers' surplus occurs when the price multiplier changes from 0.9 to 0.8. The MSR in Fig. 8(a) further demonstrates an abrupt change of the solution region: only one zone (Zone 51 which has the highest base demand) has the MSR below 1/3 when the price multiplier is set at 0.9 while all zones experience an MSR below 1/3 when the price multiplier is set at 0.8. Facing the uncertainties in marketplace, the platform may rather set a higher price to avoid an unanticipated demand surge if price is not adjustable (Castillo et al., 2017).

7. Commission rate cap regulation

Despite the flexibility offered by spatial pricing, customers may be made worse off if price is set exceptionally high. Below we present a simple regulation scheme that reaps the flexibility of spatial pricing while aiming to protect customers. We first discuss the proposed regulation and then demonstrate its performance using numerical experiments.

7.1. Regulation

Zha et al. (2016) show that regulating the amount of commission alone could achieve the second best using an aggregate and static model. With a properly chosen commission cap, a revenue-maximizing platform has the incentive to maximize the number of transactions which is positively related to consumers' surplus. The additional revenue from the price surge goes completely to drivers.

Similarly, the key idea for managing spatial pricing (or more generally surge pricing) from a public perspective is to align the platform's incentive of revenue maximization with the goal of protecting the welfare of the customers. Given that trip duration varies in the spatial network, we propose to cap the commission rate charged per unit time. A similar commission rate cap can be jointly applied for the distance-based charge for pratical implementation.

Under the proposed commission rate regulation, the commission rate is independent of price surge and the platform's revenue comes from the total occupied vehicle hours. The former solves the public's suspicion of the platform's price gouging for its own benefit. In addition, the proposed regulation shares the spirit of efficient pricing. For example, when price surge is triggered by a systematic supply shortage (e.g., on the New Year's Eve or on a chilly morning after a storm), it is likely that the cost of supplying work hours gets higher. Drivers should thus be compensated more. As the marginal operation cost of the ride-sourcing platform does not vary too much whether surge is on or not, so should be its compensation. A commission schedule independent to price surge better serves this purpose.

Mathematically, the long-run market equilibrium under the proposed regulation is given by solving the following problem:

(R-L)

$$\max_{\mathbf{N},\mathbf{z},\eta_v} = \sum_i \eta_v l_i Q_i$$
s.t.
$$\eta_v \leqslant \bar{\eta} \tag{24}$$

$$(F_i - \eta_v l_i) Q_i = cN_i\,, \quad \forall\, i \in I$$
$$\text{Eqs.(1),(14),(17)} \tag{25}$$

where η_v and $\bar{\eta}$ are respectively the commission rate and the commission rate cap per unit time. A direct application of the Envelope Theorem yields that the optimal value of R-L is non-decreasing in $\bar{\eta}$. It is thus reasonable to assume that Eq. (24) is binding for a certain range of $\bar{\eta}$. The optimal solution of R-L essentially maximizes the total occupied vehicle hours.

To evaluate the performance of the proposed regulation, we rely on a second-best scenario as the benchmark which maximizes the social welfare subject to the spatial equilibrium constraints:

(SB-L)

$$\max_{\mathbf{N},\mathbf{z},\eta_v} = \sum_i \left(\int_0^{Q_i} f^{-1}(z)dz - cN_i - \beta Q_i w_i^c\right)$$
s.t.
Eqs.(1),(14),(17)and(25)

The effectiveness of the commission cap regulation is guaranteed by the following proposition.

Proposition 3. Let λ_i, $\forall\, i \in I$ be the Lagrangian multiplier associated with Eq. (25). If $\lambda_i \neq 1$, $\forall\, i \in I$ and the commission rate is bound at the second-best level, the optimality condition for R-L is the same as that in SB-L.

Proposition 3 implies that if we choose the right commission rate cap, regulating commission rate alone can achieve the second best in a market where spatial pricing is allowed. One can verify that this result does not hold if we regulate other alternatives such as trip fare or fleet size. Therefore, regulating commission rate is an efficient and practically implementable option. The assumption on $\lambda_i \neq 1$, $\forall\, i \in I$ is due to the equal profitability assumption and will not arise in the setting of an aggregate model. Similar to Zha et al. (2016), the assumption of the uniform values of waiting time and the opportunity cost plays a key role in the proof, the relaxation of which greatly complicates the discussion of the regulation policy (Beesley and Glaister, 1983; Zha et al., 2016).

7.2. Numerical experiments

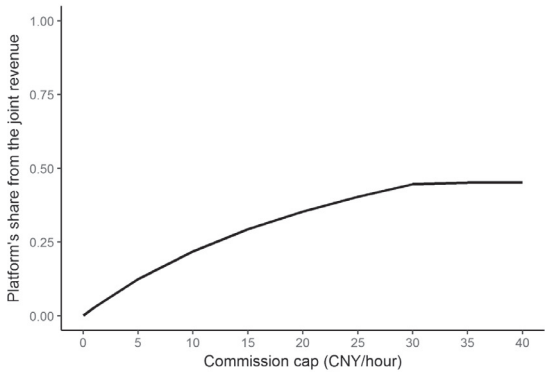
The equilibrium hourly wage rate and values of θ_1 and θ_2 are the same as those in Section 6.1. Note that when the commission cap goes to infinity, the platform becomes an unregulated monopoly with an hourly commission rate of CNY 30.77. The commission rate will be binding at this value for any larger cap.

For a commission cap less than 30.77 CNY/h, the increase in the cap corresponds to an increase in a platform's revenue split, as illustrated in Fig. 9(a). Fig. 9(b) reports the ratio of the total social surplus to that under a second-best scenario. The commission rate at the second best is zero as the operation cost for the platform is not specified in our study. Therefore, we match it with an arbitrarily small commission rate (0.001 CNY/h) to retain a legitimate objective function to optimize. The resulting social welfare accounts for over 99.99% of that under the second best. In another extreme, the social surplus of the unregulated monopoly is only 77.51% of that under the second best. The proposed regulation improves market efficiency for a wide range of commission rate caps. The platform is able to exploit drivers with high commission rates, which, as shown in Fig. 9(c), results in a reduction of the fleet size in the long term.

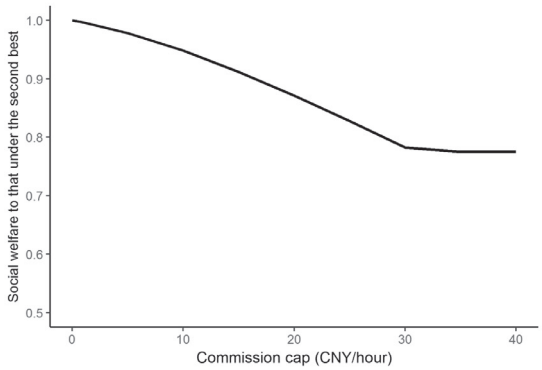
In general, the proposed regulation seeks to balance the revenue between the platform and the welfare of its users. However, the proper choice of the commission cap for practical implementation should fully consider the cost components of the platform and perhaps a reasonable profit margin. As previously mentioned, one of the core features of Didi's dual cap control is the commitment that the entire extra surged fare is awarded to drivers. A similar commission rate control is then available. Given that Didi charges a flat 20% from each transaction, it results in a normalized rate of CNY 16.2 out of 81 per hour. If a 50% margin is allowed, the corresponding commission rate cap would be CNY 24.3 per hour. From Fig. 9(b), the corresponding social welfare will be approximately 85% of that under the second best, which is still much better than that of an unregulated monopoly.

8. Conclusion

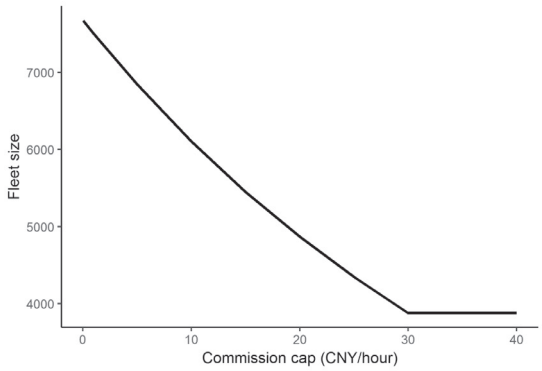
This paper investigates the effects of spatial pricing and its regulation in ride-sourcing markets. Our model is built upon a geometric matching framework, where a customer is assumed to be matched with her closest vehicle within a matching radius. Compared with street hailing, the extended matching distance reduces information imperfection but may still lead to inefficient matching. When vehicle supply is in shortage, a platform may match a customer to a distant vehicle, although the vehicle may turn out to be closer to another customer. This leads to a situation where a substantial portion of vehicle hours is spent on en route trips rather than effectively clearing the demand, eventually driving the market into an inefficient state. A tipping point for the change of the market state can be analytically identified when the number of idel vehicles is smaller than a certain percentage of the total empty or vacant vehicles in a geographic area.



(a) Share of platform's revenue



(b) Change in total social welfare



(c) Fleet size

Fig. 9. Equilibrium outcomes under the commission rate cap regulation.

Price should be increased to ration travel demand when supply is in shortage. A flat price rate that works for certain zones may turn out too high/low for others. On the other hand, spatial pricing is more flexible as it allows for a zonal-specific control. However, a revenue-maximizing platform is not likely to set price merely to balance demand and supply. Prices may be set higher for more revenue gain. We find that the platform and drivers are better off under revenue-maximizing spatial pricing while the effects on customers are ambiguous. In zones where the price multiplier is higher than that under flat pricing, we observe that customers are worse off even though the solutions are in the dominating states.

Faced with the ambiguity of spatial pricing on the welfare of customers, we propose a commission rate regulation to protect customers. It deviates from the current percentage commission structure by capping the amount of commission the platform can take from each unit time (or distance). It thus resolves the concern that the platform may manipulate price to exploit customers. The proposed regulation aligns the platform's incentive for revenue maximization with maximizing the occupied vehicle hours, which generally benefits customers; further, the additional revenue from price surge goes to drivers. With some homogeneity assumption on

agents' valuation, we show that such a simple regulation can achieve an ideal second best.

Future studies can be done in at least the following directions. First, it would be interesting to investigate the proposed matching policy via relaxing the intra-zone matching assumption to better capture the spatial heterogeneity of supply and demand. Secondly, it is necessary to empirically estimate the waiting time functions for both ride-sourcing and street-hailing technologies and then investigate their interaction and competition. Finally, a more realistic zonal choice model that accounts for the transition costs of ride-sourcing drivers is worth further investigation.

Acknowledgements

The first author appreciates earlier discussions with Kai Yin from HomeAway, Inc. on geometric matching. The work described in this paper was partly supported by the National Science Foundation – United States (CMMI-1724168 and CMMI-1740865).

Appendix A

Proof for Proposition 1

Given Eqs. (7) and (12), Eq. (4) can be spelled out as:

$$w_i^v = w_i^{vm} + \frac{k}{2v(Q_i w_i^{vm}/A_i)^\alpha}, \quad \forall\, i \in I \tag{26}$$

which essentially specifies w_i^v as a function of w_i^{vm} for a fixed demand level. One can further verify that w_i^v is convex in w_i^{vm} as

$$w_i^{v'} = 1 - \frac{k\alpha (w_i^{vm})^{-\alpha-1}}{2v(Q_i/A_i)^\alpha}, \quad \forall\, i \in I \tag{27}$$

$$w_i^{v''} = \frac{\alpha(\alpha+1)}{2}\frac{k(w_i^{vm})^{-\alpha-2}}{v(Q_i/A_i)^\alpha} > 0, \quad \forall\, i \in I \tag{28}$$

We denote w_i^{vm*} the point when w_i^v achieves its minimum w_i^{v*}. Setting $w_i^{v'} = 0$, we have $w_i^{vm*} = \frac{\alpha}{1+\alpha} w_i^{v*}$ and $w_i^{v*} = \frac{1+\alpha}{\alpha}\left[\frac{k\alpha}{2v(Q_i/A)^\alpha}\right]^{\frac{1}{1+\alpha}}$. Therefore case (i) holds. The latter two cases follow naturally since $\lim_{w_i^{vm}\to\infty} w_i^v = \infty$ and $\lim_{w_i^{vm}\to\delta} w_i^v < \infty$.

Proof for Lemma 1

We note that $Q_{i(A_L)} = Q_{i(A_R)}$, $w_{i(A_L)}^v = w_{i(A_R)}^v$ by construction. It is straightforward to see $w_{i(A_L)}^{vm} < w_{i(A_R)}^{vm}$, which immediately gives $w_{i(A_L)}^{cm} > w_{i(A_R)}^{cm}$ since w_i^{vm} decreases in w_i^{cm}. By Little's law, $N_{i(A_L)}^{vm} = Q_{i(A_L)} w_{i(A_L)}^{vm}/A_i < Q_{i(A_R)} w_{i(A_R)}^{vm}/A_i = N_{i(A_R)}^{vm}$. Therefore, $m_{i(A_L)} > m_{i(A_R)}$ and $w_{i(A_L)}^c > w_{i(A_R)}^c$.

As demand is decreasing in total cost, $Q_{i(A_L)} = Q_{i(A_R)}$ implies $F_{i(A_L)} + \beta w_{i(A_L)}^c = F_{i(A_R)} + \beta w_{i(A_R)}^c$. Since we have shown $w_{i(A_L)}^c > w_{i(A_R)}^c$, it must be that $F_{i(A_L)} < F_{i(A_R)}$. As will be shown in Eq. (17), $N_i = Q_i(w_i^v + l_i)$ must hold for vehicle status conservation in a steady state, which leads to $N_{i(A_L)} = N_{i(A_R)}$. By checking the specific components in Eq. (13) and utilizing the relationships that have already been established, we claim that $CS_{i(A_L)} = CS_{i(A_R)}$, $PS_{i(A_L)} < PS_{i(A_R)}$.

Proof for Lemma 2

The following two equalities are obtained via differentiating Eqs. (3), (4), (6) and (12):

$$W_2 = (\xi_i^{-1} + m_i' Q_i)(1 + m_i' Q_i)^{-1}, \quad \forall\, i \in I \tag{29}$$

$$W_1 = m_i' w_i^{vm}(1 - W_2), \quad \forall\, i \in I \tag{30}$$

where $\xi_i = -\ln(1-\delta/w_i^{cm}) - \delta/(w_i^{vm} - \delta) < 0$ and $m_i' = -\alpha m_i/(Q_i w_i^{vm}) < 0$. The key component to evaluate the signs of W_1, W_2 is $1 + m_i' Q$. It is obvious that $1 + m_i' Q_i > 0$ if and only if $w_i^{vm} > \frac{\alpha}{1+\alpha} w_i^v$; $1 + m_i' Q_i < 0$ if and only if $w_i^{vm} < \frac{\alpha}{1+\alpha} w_i^v$. In the former case, $W_1 < 0$, $W_2 < 0$. In the latter case, $W_2 > m_i' Q_i/(1 + m_i' Q_i) > 1$ and by Eq. (30) we obtain that $W_1 > 0$.

Proof for Proposition 2

The general idea is to prove by contradiction. If the optimal solution is in the dominated region, we will then find another point in the dominating region such that the platform's revenue is non-decreasing. More specifically, we will prove the existence of an equilibrium point in the dominating region with $\Delta F_i > 0$, $\Delta\eta = 0$ and $\Delta Q_i = 0$.

The requirement on $\Delta\eta = 0$ implies that the changes of w_i^v and F_i in Eq. (18) must satisfy:

$$\Delta w_i^v = \frac{(1-\eta)}{c}\Delta F_i, \quad \forall\, i \in I \tag{31}$$

Give that $\Delta Q_i = 0$, the total travel cost remains unchanged. Since $\Delta F_i > 0$, we have:

$$\Delta w_i^c = -\beta/\Delta F_i < 0\ , \quad \forall\ i \in I \tag{32}$$

where Δw_i^c is related to Δw_i^v via $\Delta w_i^c = W(Q_i,w_i^v + \Delta w_i^v)_R - W(Q_i,w_i^v)_L$. Since we have assumed an optimal solution in the dominated region, we note from Proposition 1 that $W(Q_i,w_i^v + \Delta w_i^v)_R$ and $W(Q_i,w_i^v)_L$ must exist. Then, Δw_i^v can be solved from the following equation:

$$\Delta w_i^v = \frac{(1-\eta)\beta}{c}[W(Q_i,w_i^v)_L - W(Q_i,w_i^v + \Delta w_i^v)_R] \quad \forall\ i \in I \tag{33}$$

Let's define $T(x) = [W(Q_i,w_i^v)_L - W(Q_i,w_i^v + x)_R](1-\eta)\beta/c - x$. We note that $\lim_{x\to\infty} T(x) = -\infty$ and $\lim_{x\to 0} T(x) = [W(Q_i,w_i^v)_L - W(Q_i,w_i^v)_R](1-\eta)\beta/c > 0$. As $W(Q_i,w_i^v)_L$ and $W(Q_i,w_i^v)_R$ are assumed to vary continuously w.r.t. w_i^v in the dominating and dominated region, Eq. (33) admits at least one solution.

Finally, it is trivial to see that $\eta(F_i + \Delta F_i)(Q_i + \Delta Q_i) - \eta F_i Q_i > 0$, which conflicts with the assumption that the revenue maximizing point is in the dominated region.

Proof for Proposition 3

We first notice that Eqs. (1), (14) and (17) have $5|I|$ variables. We choose F_i and N_i as the independent variables. It is easy to show that:

$$\frac{\partial Q_i}{\partial F_i} = \frac{f_i'}{1-\beta f_i'\Delta_i}, \quad \frac{\partial Q_i}{\partial N_i} = \frac{\partial Q_i}{\partial F_i}\frac{\beta W_2}{Q_i}\ , \quad \forall\ i \in I \tag{34}$$

where $\Delta_i = W_1 - W_2(w_i^v + l_i)/Q_i$ and $1-\beta f_i'\Delta_i \neq 0$.

(i) The optimality condition of the regulated problem

The Lagrangian function of R-L is:

$$L(\mathbf{F},\mathbf{N},\boldsymbol{\lambda},\eta_v) = \sum_i \eta_v l_i Q_i - \sum_i [(F_i - \eta_v l_i)Q_i - cN_i]\lambda_i \tag{35}$$

We assume that $F_i > 0$ and $N_i > 0$ and focus on the interior solution. As we have fixed η_v at the second-best level, the first order necessary condition (FONC) of R-L is characterized by:

$$\frac{\partial Q_i}{\partial F_i}[\eta_v l_i - (F_i - \eta_v l_i)\lambda_i] = \lambda_i Q_i\ , \quad \forall\ i \in I \tag{36}$$

$$\frac{\partial Q_i}{\partial N_i}[\eta_v l_i - (F_i - \eta_v l_i)\lambda_i] = -c\lambda_i\ , \quad \forall\ i \in I \tag{37}$$

Eq. (25)

Note that $\eta_v l_i - (F_i - \eta_v l_i)\lambda_i \neq 0$ and $\lambda_i \neq 0\ , \quad \forall\ i \in I$. Otherwise it leads to $\partial Q_i/\partial F_i = 0$ and $\partial Q_i/\partial N_i = 0$ which is clearly a conflict as $f_i' < 0$ and $1-\beta f_i'\Delta_i \neq 0$. Combining Eqs. (36) and (37), we obtain that:

$$\frac{\partial Q_i}{\partial F_i} \Big/ \frac{\partial Q_i}{\partial N_i} = -\frac{Q_i}{c}\ , \quad \forall\ i \in I \tag{38}$$

Substituting $\partial Q_i/\partial F_i$ and $\partial Q_i/\partial N_i$ from Eq. (34) into Eq. (38), we have:

$$c + \beta W_2 = 0\ , \quad \forall\ i \in I \tag{39}$$

In summary, the optimality condition of R-L is characterized by Eqs. (1), (14), (17), (25) and (39).

(ii) The optimality condition of the second best

The corresponding Lagrangian function for SB-L is:

$$L(\mathbf{F},\mathbf{N},\boldsymbol{\lambda},\eta_v) = \sum_i \left(\int_0^{Q_i} f^{-1}(z)dz - cN_i - \beta Q_i w_i^c\right) - \sum_i ((F_i - \eta_v l_i)Q_i - cN_i)\lambda_i \tag{40}$$

We assume that $F_i > 0$, $\eta_v > 0$ and $N_i > 0$. Its FONC is given by:

$$\frac{\partial Q_i}{\partial F_i}[F_i - (F_i - \eta_v l_i)\lambda_i - \beta Q_i\Delta_i] = \lambda_i Q_i\ , \quad \forall\ i \in I \tag{41}$$

$$\frac{\partial Q_i}{\partial N_i}[F - (F_i - \eta_v l_i)\lambda_i - \beta Q_i\Delta_i] = (1-\lambda_i)c + \beta W_2\ , \quad \forall\ i \in I \tag{42}$$

$$\sum_i Q_i l_i \lambda_i = 0 \tag{43}$$

Eq. (25)

(Case 1)

If $\exists\ i \in I$, $\lambda_i = 0$, it is immediate from Eqs. (41) and (42) that $c + \beta W_2 = 0$ for such zones. However, there must be some zone $j \in I$ where $\lambda_j \neq 0$. Otherwise, the industry is not sustainable as $\sum (F_i Q_i - cN_i) = \sum \beta Q_i^2 W_1 < 0$. The negativity is simply because the optimal solution must be in the dominating region. One can follow the same exercise as we have done in Proposition 2 to prove this. In fact, $\lambda_i = 0\ ,\quad \forall\ i \in I$ corresponds to the first-best scenario which needs to be subsidized (e.g., Douglas, 1972; Arnott, 1996; Yang et al., 2010b; Zha et al., 2016).

(Case 2)

We next focus on the zones with $\lambda_j \neq 0$, $j \in I$. Dividing Eq. (41) by Eq. (42), we obtain that:

$$\frac{\partial Q_i}{\partial F_i} \Big/ \frac{\partial Q_i}{\partial N_i} = \frac{\lambda_i Q_i}{(1-\lambda_i)c + \beta W_2}\ ,\quad \forall\ i \in I \cap \{j | \lambda_{j \in I} \neq 0\} \tag{44}$$

Substituting $\partial Q_i/\partial F_i$ and $\partial Q_i/\partial N_i$ from Eq. (34) into Eq. (44), we have:

$$(1-\lambda_i)(\beta W_2 + c) = 0\ ,\quad \forall\ i \in I \cap \{j | \lambda_{j \in I} \neq 0\} \tag{45}$$

As we have imposed the assumption that $\lambda_i \neq 1\ ,\quad \forall\ i \in I$. It must be that $\beta W_2 + c = 0$, $\forall\ i \in I \cap \{j | \lambda_{j \in I} \neq 0\}$.

Put the discussions in Case 1 and 2 together, we claim that Eq. (39) holds when $\lambda_i \neq 1\ ,\quad \forall\ i \in I$. In summary, the optimality condition under the second best is characterized by Eqs. (1), (14), (17), (25) and (39), which is the same as those from R-L.

Appendix B

We define y_{i-} the rate of existent unmatched vehicles that move to zone i. Then the equal profitability condition can be generalized to:

$$\left[(1-\eta)\frac{F_i Q_i}{N_i} - \pi\right] y_{i-} = 0 \quad (1-\eta)\frac{F_i Q_i}{N_i} - \pi \leqslant 0\ ,\quad \forall\ i \in I \tag{46}$$

With some abuse of notation, π is now interpreted as the maximum equilibrium revenue. The complementarity formulation implies that $y_{i-} = 0$ if $(1-\eta)F_i Q_i/N_i - \pi < 0$, or unmatched drivers will not enter less attractive zones; $y_{i-} \geqslant 0$ if $(1-\eta)F_i Q_i/N_i - \pi = 0$, or they will be attracted to more promising zones. Eq. (16) only considers the latter case.

The quantification of y_{i-} requires the explicit description of the movement of vehicles. We assume that the transition matrix of customers Λ is given where Λ_{ij} is the proportion of customers who wish to move from zone i to zone j. Given that Q_i is the rate of demand originating from zone i, the rate of demand ending at zone i is then $\sum_{j \in I} Q_j \Lambda_{ji}$.

Next, consider the flow rate conservation in the spatial market:

$$T_i^v = \sum_{j \in I} Q_j \Lambda_{ji} + y_{i-}\ ,\quad \forall\ i \in I \tag{47}$$

$$T_i^v = Q_i + y_{-i}\ ,\quad \forall\ i \in I \tag{48}$$

where y_{-i} represents the rate of unmatched vehicles that remain unmatched after searching in zone i; T_i^v is the total rate of unmatched vehicles originating from zone i.

So far, we have introduced additional 3 |I| variables but only 2 |I| equations. One can take the indeterminate nonlinear equation system and study the constrained pricing problems (Bimpikis et al., 2016). This provides an ideal bound for the true market outcomes. Although the enhanced formulation is flexible to capture the asymmetric demand patterns, it comes with the price that the model becomes intractable. Special algorithms need to be devised to solve the pricing problems with the complementarity constraints.

In general, spatial modeling of ride-for-hire networks is a challenging task. Among other endeavors, a large body of existing literature follows a transit-and-search approach developed by Yang and Wong (1998). An empty vehicle is not allowed to be matched during transit and is assumed to search in the chosen transition destination until picking up a customer. Wong et al. (2014) enhance the model by capturing the sequential search process using an intervening opportunity model. However, the model is not tractable to be integrated into our geometric matching framework.

References

Arnott, R., 1996. Taxi travel should be subsidized. J. Urban Econ. 40, 316–333.

Banerjee, S., Johari, R., Riquelme, C., 2015. Pricing in ride-sharing platforms: a queueing-theoretic approach. In: Proceedings of the Sixteenth ACM Conference on Economics and Computation. ACM, pp. 639–639.

Beesley, M.E., Glaister, S., 1983. Information for regulating: the case of taxis. Econ. J. 93, 594–615.

Bimpikis, K., Candogan, O., Daniela, S., 2016. Spatial pricing in ride-sharing networks. Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2868080. Accessed Jul., 20, 2017.

Cachon, G.P., Daniels, K.M., Lobel, R., 2015. The role of surge pricing on a service platform with self-scheduling capacity. Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2698192. Accessed Dec., 30, 2015.
Castillo, J.C., Knoepfle, D., Weyl, E.G., 2017. Surge pricing solves the wild goose chase. Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2890666. Accessed Sep., 20, 2017.
Chen, L., Mislove, A., Wilson, C., 2015. Peeking beneath the hood of Uber. In: Proceedings of the 2015 ACM Conference on Internet Measurement Conference, ACM, pp. 495–508.
Chen, M.K., Sheldon, M., 2016. Dynamic Pricing in A Labor Market: Surge Pricing and Flexible Work on the Uber Platform, UCLA Anderson School of Management.
Chiu, S.N., Stoyan, D., Kendall, W.S., Mecke, J., 2013. Stochastic Geometry and Its Applications. Wiley.
Curley, B., 2014. A deep look at Uber's dynamic pricing model. http://abovethecrowd.com/2014/03/11/a-deeper-look-at-ubers-dynamic-pricing-model/. Accessed Nov. 17, 2014.
Didi Chuxing, 2016a. Didi algorithm competition. http://research.xiaojukeji.com/competition/main.action?competitionId=DiTech2016. Accessed Jan. 14, 2017.
Didi Chuxing, 2016b. Report on Residents' Traveling Data: Case of Hang Zhou. http://www.useit.com.cn/thread-12885-1-1.html. Accessed Feb., 18, 2017.
Didi Chuxing, 2017. A response to liuliu. http://finance.sina.com.cn/chanjing/gsnews/2017-06-22/doc-ifyhmtcf2768104.shtml. Accessed July 20, 2017.
Douglas, G.W., 1972. Price regulation and optimal service standards: the taxicab industry. J. Transport Econ. Policy 6, 116–127.
Farber, H.S., 2015. Why you can't find a taxi in the rain and other labor supply lessons from cab drivers. Q. J. Econ. 130, 1975–2026.
Galichon, A., Hsieh, Y.-W., 2017. A theory of decentralized matching markets without transfers, with an application to surge pricing. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2908532. Accessed Aug., 28, 2017.
Guda, H., Subramanian, U., 2017. Strategic surge pricing and forecast communication on on-demand service platforms. Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2895227. Accessed Aug., 28, 2017.
Hall, J., Kendrick, C., Nosko, C., 2015. The effects of Uber's surge pricing: a case study, uber technologies.
Hall, J.V., Krueger, A.B., 2015. An Analysis of the Labor Market for Uber's Driver-Partners in the United States, Princeton University Industrial Relations Section.
Hansen, B., 2017. Econometrics. University of Wisconsin.
Hu, M., Zhou, Y., 2017. Price, Wage and Fixed Commission in On-Demand Matching, Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2949513. Accessed Aug., 28, 2017.
Lagos, R., 2000. An alternative approach to search frictions. J. Political Econ. 108, 851–873.
Moltchanov, D., 2012. Distance distributions in random networks. Ad Hoc Netw. 10, 1146–1166.
Nicholas, C., 2016. Experimentation in A Ridesharing Marketplace. https://eng.lyft.com/experimentation-in-a-ridesharing-marketplace-b39db027a66e. Accessed Jul., 20, 2017.
Nie, Y., 2017. How can the taxi industry survive the tide of ridesourcing? Evidence from Shenzhen, China. Transportation Res. Part C: Emerg. Technol. 79, 242–256.
Ranney, M., 2015. Scaling Uber's real-time market platform. QCON London. http://www.slideshare.net/InfoQ/scaling-ubers-realtime-market-platform-50758747. Accessed Nov., 23, 2015.
Taylor, T., 2016. On-demand service platforms. Available at SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2722308. Accessed Dec., 30, 2016.
Vincent, L.-M., 2016. When to quit: narrow bracketing and reference dependence in taxi drivers. Working Paper at the University of California, San Diego.
Wong, R., Szeto, W., Wong, S., 2014. A cell-based logit-opportunity taxi customer-search model. Transportation Res. Part C: Emerg. Technol. 48, 84–96.
Xu, Z., Yin, Y., Zha, L., 2017. Optimal parking provision for ride-sourcing services. Transportation Res. Part B: Methodol. 105, 559–578.
Yang, H., Fung, C.S., Wong, K.I., Wong, S.C., 2010a. Nonlinear pricing of taxi services. Transportation Res. Part A: Policy Practice 44, 337–348.
Yang, H., Leung, C.W.Y., Wong, S.C., Bell, M.G.H., 2010b. Equilibria of bilateral taxi-customer searching and meeting on networks. Transportation Res. Part B: Methodol. 44, 1067–1083.
Yang, H., Wong, S.C., 1998. A network model of urban taxi services. Transportation Res. Part B: Methodol. 32, 235–246.
Yang, H., Ye, M., Tang, W.H.-C., Wong, S.C., 2005. A multiperiod dynamic model of taxi services with endogenous service intensity. Oper. Res. 53, 501–515.
Zha, L., Yin, Y., Du, Y., 2017. Surge pricing and labor supply in the ride-sourcing market. Transportation Res. Part B: Methodol. http://dx.doi.org/10.1016/j.trb.2017.09.010.
Zha, L., Yin, Y., Yang, H., 2016. Economic analysis of ride-sourcing markets. Transportation Res. Part C: Emerg. Technol. 71, 249–266.